

SLoMO Technical Report

Movie-Level Joint Inference for Long-Video QA

Reproducibility documentation and implementation notes

System Overview

The system answers open-ended questions about short films. Its basic unit is the movie rather than the individual question: all questions for one film are answered in a shared multimodal context. This preserves information that is otherwise difficult to recover independently, including aliases, changing relationships, event order, causes, and the ending state.

The artifact releases the executable inference and submission utilities, configuration guidance, tests, chart data, and this report. Videos, subtitles, model weights, generated answers, credentials, account identifiers, and evaluation records remain external inputs.

Method

For movie m , let E_m denote the shared evidence and Q_m its complete question set. The runner makes one request and receives one answer bank:

$$(E_m, Q_m) \longrightarrow \{a_{m,1}, a_{m,2}, \dots, a_{m,n_m}\}.$$

The evidence packet contains a timestamped transcript and a chronological frame pack. The prompt asks for concise, evidence-supported English answers with exact question IDs. It requires the model to maintain cross-question consistency and forbids unsupported visual details.

Frames are sampled in chronological order, resized to a fixed maximum side, JPEG encoded, and marked with their timestamps. Long transcripts use a deterministic head-and-tail bound. The model is accessed through an OpenAI-compatible Responses endpoint; the credential is read from an environment variable at runtime.

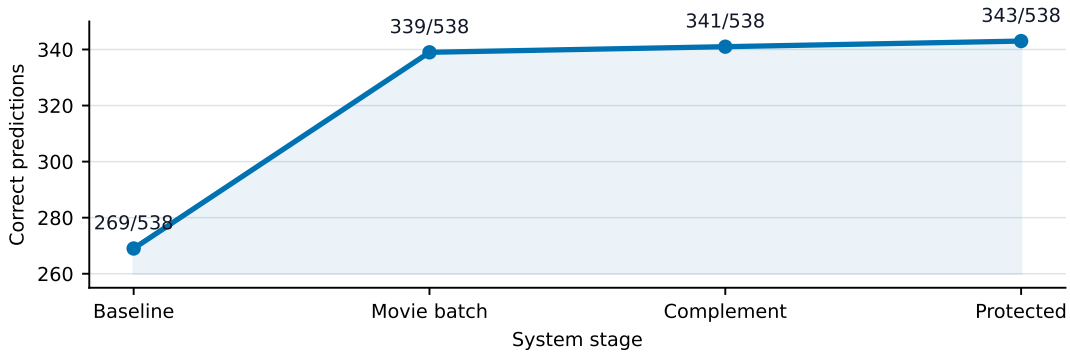


Figure 1: Historical official score chain recorded in the project audit. The large discontinuity coincides with the movie-batched multimodal answer source; later parent-preserving changes are smaller.

The chart is a historical reference, not a claim that the released artifact reproduces every upstream service state. It motivates the movie-level answer bank interface used by the runner.

Inference Pipeline

The deterministic pipeline groups rows by `video_id`, parses timestamped subtitle cues, samples chronological frames, sends one request with all movie questions, and validates the complete JSON bank before persistence. The final conversion writes the competition CSV. Missing IDs, empty answers, malformed JSON, and incomplete movie banks are errors; partial responses are never merged.

Reproducibility

Component	Fixed setting
Context unit	One movie, all questions jointly
Evidence	Timestamped transcript + 16 chronological JPEG frames
Frame encoding	Maximum side 512 px; JPEG quality 82
Generation	Temperature 0; high reasoning effort
Output	JSON answer bank; CSV <code>question_id,prediction</code>

Example execution is documented in `README.md`. The audit checks the header, exact ID coverage, uniqueness, and non-empty predictions. Public and private splits are supplied separately.

Results and Ablations

Figure 2 reports a frozen six-movie, 60-question local diagnostic using one semantic evaluator. Joint answering reaches 27/60, independent calls 18/60, no frames 24/60, Target16 29/60, and Dense48 30/60. The results support joint context and bound simple frame-budget changes.

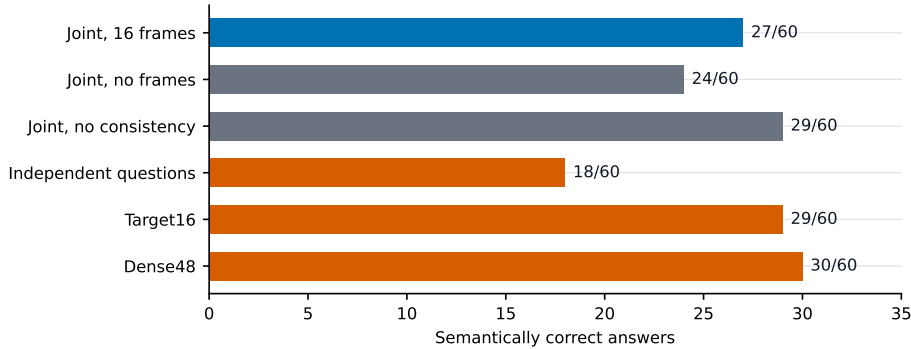


Figure 2: Frozen factor and visual-ceiling diagnostic on six movies. Blue marks the direct reference; gray and orange bars are controlled variants. These are local diagnostics, not official leaderboard scores.

Limitations

The method depends on transcript quality, visual coverage, endpoint behavior, and context budget. Data and weights are external inputs, and endpoint availability is not guaranteed. Reproduction requires authorized access and the included schema audit.

References and Public Sources

1. SLoMO Workshop and Challenge Page.
<https://slomo-workshop.github.io/eccv2026/>
2. EvalAI Evaluation Platform.
<https://eval.ai/>
3. SF20K Dataset Landing Page.
<https://huggingface.co/datasets/rghermi/sf20k>

The live competition page and evaluation service are authoritative for current deadlines, phases, metadata fields, and upload rules. The report describes the artifact interface and the recorded evidence available at release time.